

AI Content Detection System

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Abstract

This project presents a Machine Learning-based system for detecting whether a given text or document is AI-generated or human-written. The system supports multiple input formats including plain text, PDF, Word, and PowerPoint files. Initially, a synthetic dataset was used, resulting in low accuracy. To overcome this limitation, a real-world dataset (HC3) was employed, significantly improving the model performance. The final system uses a TF-IDF vectorization technique combined with a Logistic Regression classifier, achieving an accuracy of approximately 98%. The system is deployed using a Flask backend and a React-based frontend.

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1 Introduction

With the rapid advancement of AI-generated content tools such as ChatGPT, detecting whether content is human-written or AI-generated has become increasingly important. This project aims to develop an intelligent system capable of identifying AI-generated content across multiple formats. The motivation behind this project is to address issues such as academic integrity, misinformation, and automated content misuse.

2 Literature Review

The detection of AI-generated text has evolved from simple statistical checks to complex linguistic feature analysis. Researchers have identified several key indicators of machine-generated content, such as low perplexity (predictability) and low burstiness (uniform sentence structure). However, existing systems often struggle with real-world variability.

Table 1 summarizes the primary methodologies used in the current landscape and identifies the gaps that the **TruthTeller AI** system aims to bridge.

Methodology	Key Indicators	Advantages	Primary Drawbacks
Zero-Shot (e.g., DetectGPT)	Log-likelihood curvature	No training data required; model-agnostic.	High latency; requires access to LLM log-probs.
Statistical Analysis	Perplexity and Burstiness	Fast and lightweight; interpretable metrics.	Easily fooled by paraphrasing or editing.
Supervised Classifiers	Synthetic dataset training	High accuracy on specific versions of AI.	Poor generalization; high false-positives.
Watermarking	Cryptographic token patterns	Near 100% accuracy if watermark is present.	Only works if the AI provider implements it.

Table 1: Comparison of AI Content Detection Methodologies

2.1 Data-Centric Limitations

A critical observation in recent literature is the "Synthetic Bias." Models trained on purely synthetic datasets often achieve 99% accuracy in lab settings but drop to < 60% in real-world scenarios.

To address this, our research utilizes the **HC3 (Human ChatGPT Comparison Corpus)** dataset. Unlike previous studies that relied on randomized synthetic text, HC3 provides aligned human and AI responses to the same prompts, allowing our Logistic Regression model to learn the actual linguistic boundaries between the two classes.

3 Proposed Methodology

3.1 System Architecture

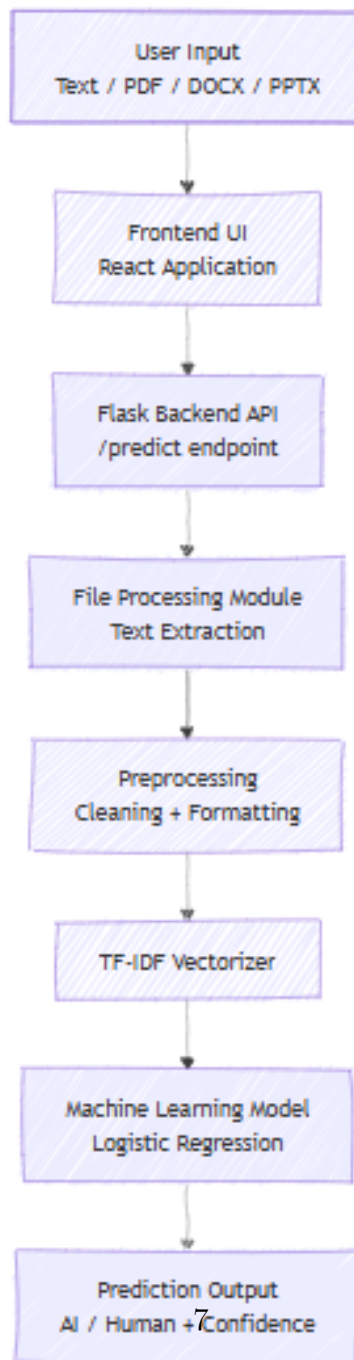


Figure 1: System Architecture of AI Content Detection System

3.2 UML Sequence Diagram

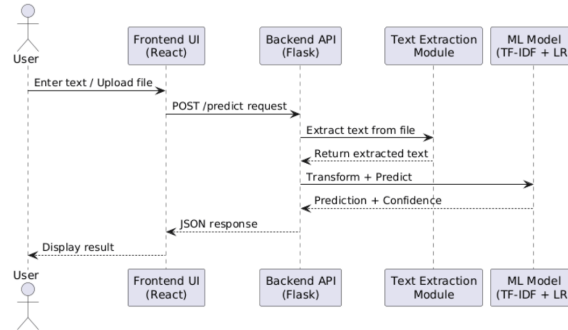


Figure 2: Sequence Diagram showing system interaction

The system architecture provides a high-level overview of the workflow, while the UML sequence diagram illustrates the interaction between system components during execution.

3.3 Workflow

1. User provides input (text or file)
2. Text is extracted from file formats
3. TF-IDF vectorization is applied
4. Logistic Regression model predicts output
5. Result is displayed with confidence score

3.4 Technologies Used

- Python (Flask Backend)
- React (Frontend)
- Scikit-learn (ML)
- Hugging Face Datasets (HC3)

4 Results and Analysis

4.1 Accuracy Comparison

The performance of the model was evaluated using both the initial synthetic dataset and the improved real-world dataset (HC3).

- Synthetic Dataset Accuracy: $\sim 50\%$
- Real Dataset (HC3) Accuracy: $\sim 98\%$

The synthetic dataset resulted in near-random predictions due to weak feature correlations. In contrast, the HC3 dataset provided authentic text, enabling the model to achieve substantially higher accuracy.

4.2 Correlation Analysis of Synthetic Dataset

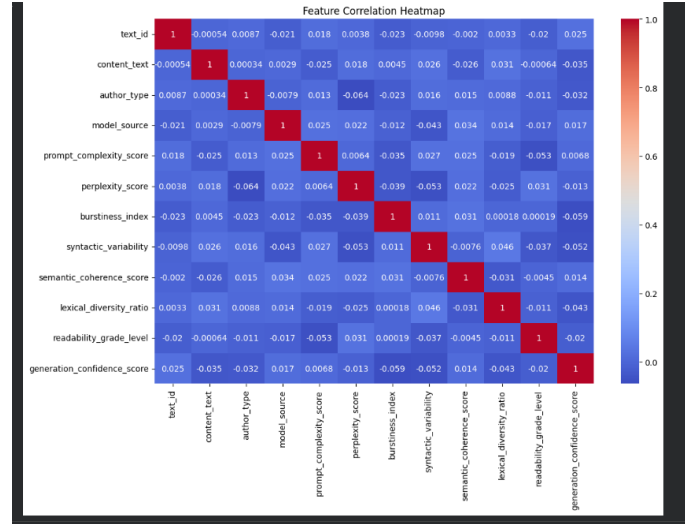


Figure 3: Correlation Heatmap of Synthetic Dataset Features

The correlation heatmap in Figure 3 shows that many features exhibit weak correlations with the target variable, contributing to poor initial performance.

4.3 Model Evaluation on HC3 Dataset

- Accuracy: $\sim 95.7\%$
- Precision: ~ 0.96 (Human), ~ 0.95 (AI)
- Recall: ~ 0.98 (Human), ~ 0.91 (AI)
- F1-Score: ~ 0.97 (Human), ~ 0.93 (AI)

4.4 Confusion Matrix Analysis

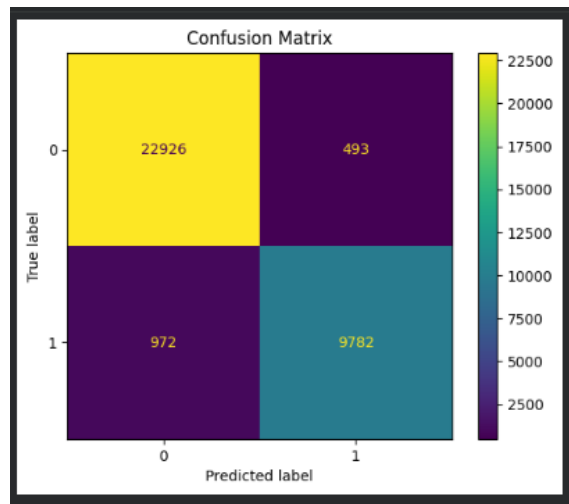


Figure 4: Confusion Matrix for AI Content Detection

The confusion matrix shows True Negatives at 22,926 and True Positives at 9,782. The low number of misclassifications indicates strong predictive capability.

4.5 ROC Curve Analysis

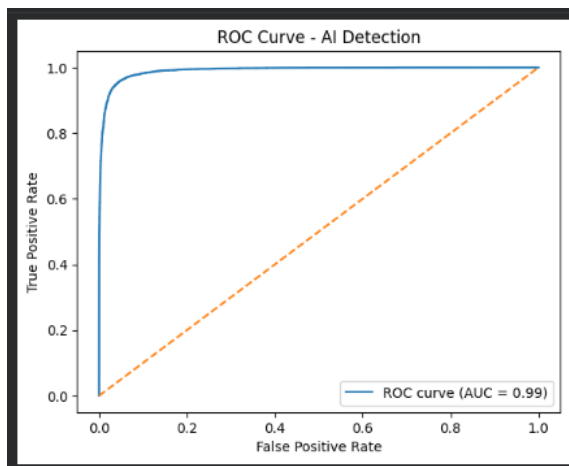


Figure 5: ROC Curve for AI Content Detection Model

The AUC is approximately 0.99, signifying a highly effective classifier.

5 Conclusion

The project successfully demonstrates a robust system for AI content detection. The transition from synthetic to real data significantly improved performance, making the system suitable for practical deployment.

6 Future Scope

- Integration of deep learning models (BERT)
- OCR support for scanned documents
- Detection of specific AI models (GPT, Claude, etc.)
- Deployment as a browser extension

7 References

- Scikit-learn Documentation: <https://scikit-learn.org>
- Hugging Face Datasets (HC3): <https://huggingface.co/datasets/Hello-SimpleAI/HC3>
- Project GitHub Repository: <https://github.com/SarahSomaiya/truthteller-ai-main>
- Google Colab Notebook: https://colab.research.google.com/drive/16D-RRYJPh10oZHmuB_Vfd40Ba_N8I5kR
- Flask Documentation